

Subtracting Compact Binary Foregrounds to Search for Subdominant Gravitational-Wave Backgrounds in Next-Generation Ground-Based Observatories

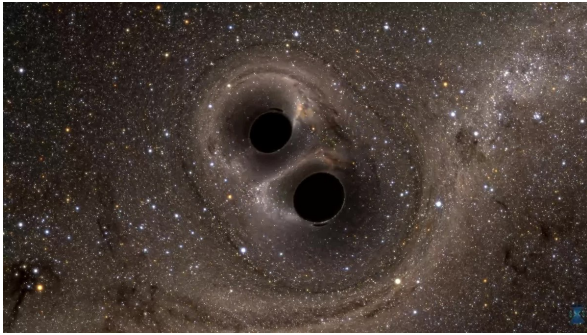
Cyril Creque-Sarbinowski

PACMAN 2022



2209.01221, 2209.01221: B. Zhou, L. Reali, E. Berti, M. Çalışkan, M. Kamionkowski, B. S. Sathyaprakash

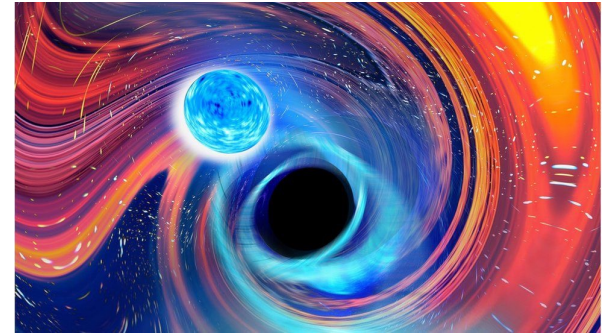
What are Compact Binaries?



Binary Black Holes (BBH)



Binary Neutron Stars (BNS)

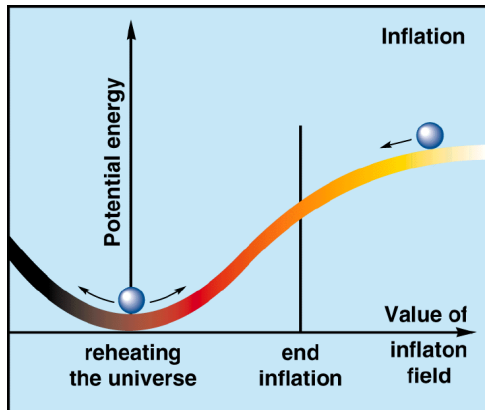


Neutron Star-Black Holes (NSBH)

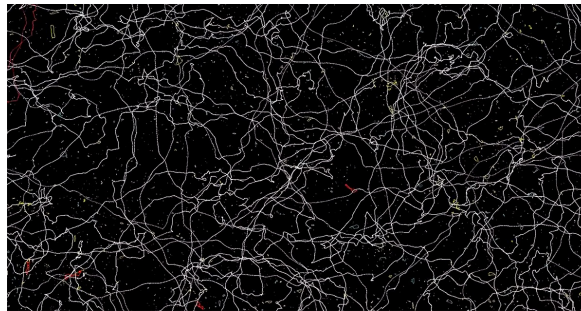
How many events have we seen?

		GWTC1 (2015-2017): 11	GWTC2.1 (Apr-Oct 2019): 44	GWTC3 (2019-2020): 35	Total: 90
BBH		10	43-44	33-35	86-89
BNS		1	0-1	0	1-2
NSBH		0	0	0-2	0-2

What are other GW Backgrounds?



Inflation and Preheating



Topological Defects
(Cosmic Strings)



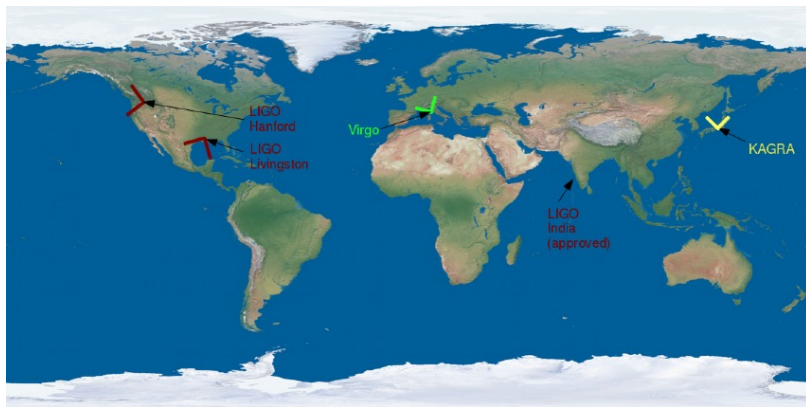
Core-Collapsed Supernovae

[Next-Generation (XG)] Ground-Based Observatories

Current

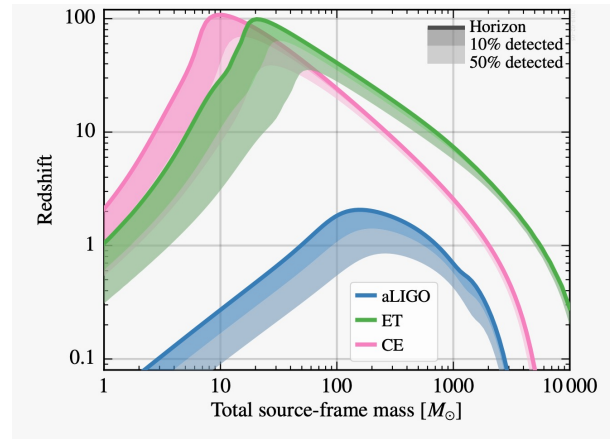
(LVK)

LIGO-Hanford VIRGO
LIGO-Livingston KAGRA

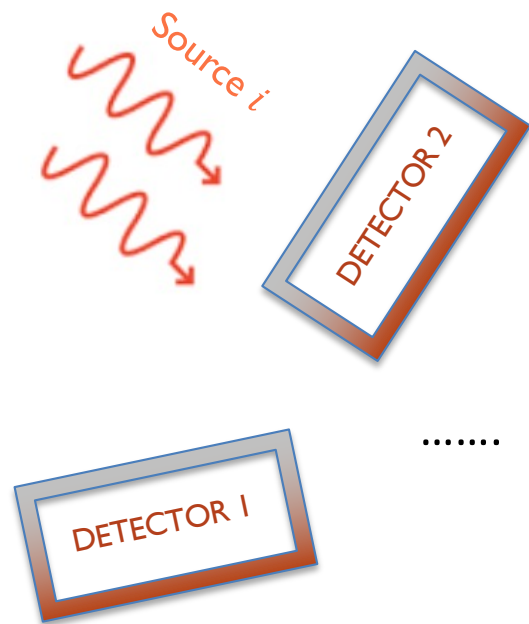


Future

Cosmic Explorer
Einstein-Telescope



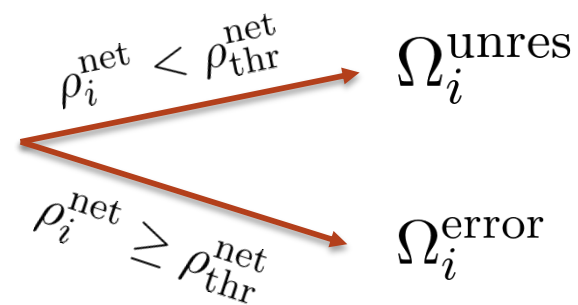
Subtraction Method



Calculate SNR

$$\rho_i^{\text{net}} = \sqrt{\sum_{\text{det}} (\rho_i^{\text{det}})^2}$$

$$(\rho_i^{\text{det}})^2 = 4 \int_0^\infty df \frac{|\tilde{h}_i^{\text{det}}(f)|^2}{S_h^{\text{det}}(f)}$$



$$\text{CBC foreground} = \Omega_{\text{unres}} + \Omega_{\text{error}}$$

Estimating the Total Foreground

$$\Omega_{\text{unres}} + \Omega_{\text{error}}$$

$$\Omega_{\text{unres}}(f) \equiv \frac{f}{\rho_c} \frac{d\rho_{\text{unres}}}{df}(f) = \frac{1}{\rho_c c} f F_{\text{unres}}(f)$$

$$F_{\text{unres}}(f) = T^{-1} \frac{\pi c^3}{2G} f^2 \sum_{i=1}^{N_{\text{unres}}} \left[|\tilde{h}_+^i(f)|^2 + |\tilde{h}_\times^i(f)|^2 \right]$$

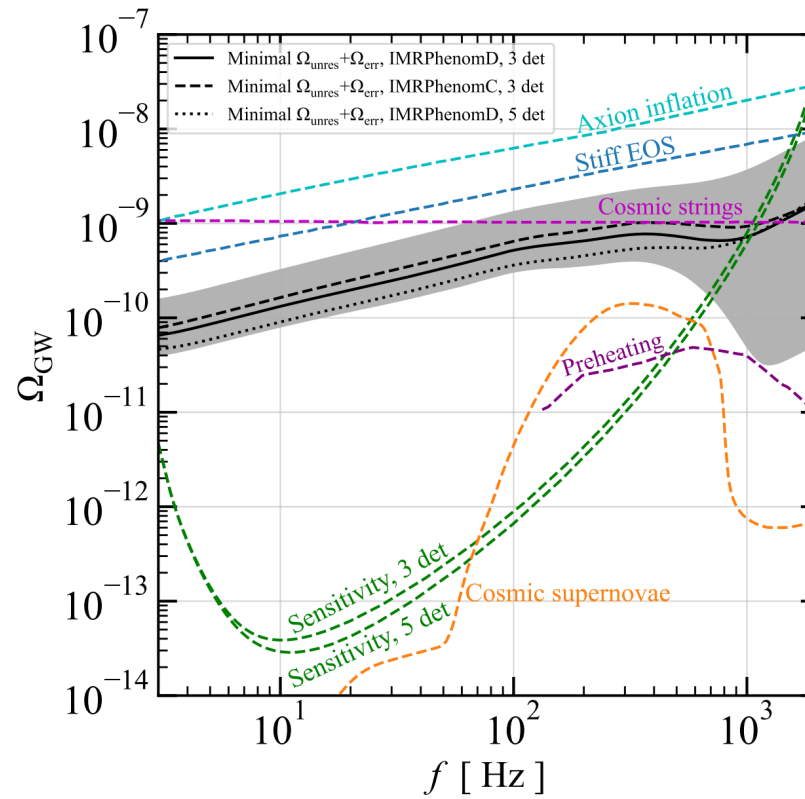
$$\Omega_{\text{error}} = \frac{1}{\rho_c c} f F_{\text{error}}(f)$$

$$F_{\text{error}}(f) = T^{-1} \frac{\pi c^3}{2G} f^2 \sum_{i=1}^{N_{\text{res}}} \left[\left| \tilde{h}_+(\boldsymbol{\theta}_{\text{tr}}^i; f) - \tilde{h}_+(\boldsymbol{\theta}_{\text{rec}}^i; f) \right|^2 + \left| \tilde{h}_\times(\boldsymbol{\theta}_{\text{tr}}^i; f) - \tilde{h}_\times(\boldsymbol{\theta}_{\text{rec}}^i; f) \right|^2 \right],$$

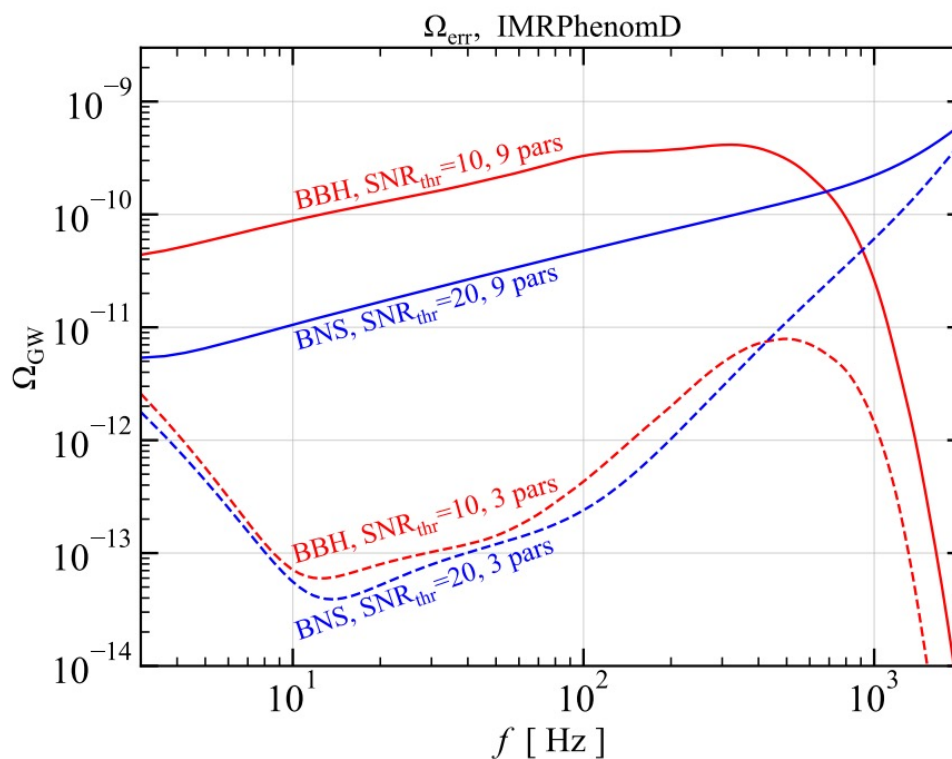
$$\boldsymbol{\theta} = \left\{ \ln\left(\frac{\mathcal{M}_z}{M_\odot}\right), \eta, \ln\left(\frac{D_L}{\text{Mpc}}\right), \cos \iota, \cos \delta, \alpha, \psi, \phi_c, t_c \right\}$$

$$N = 10^5, f \in \{3, 2000\} \text{ Hz}$$

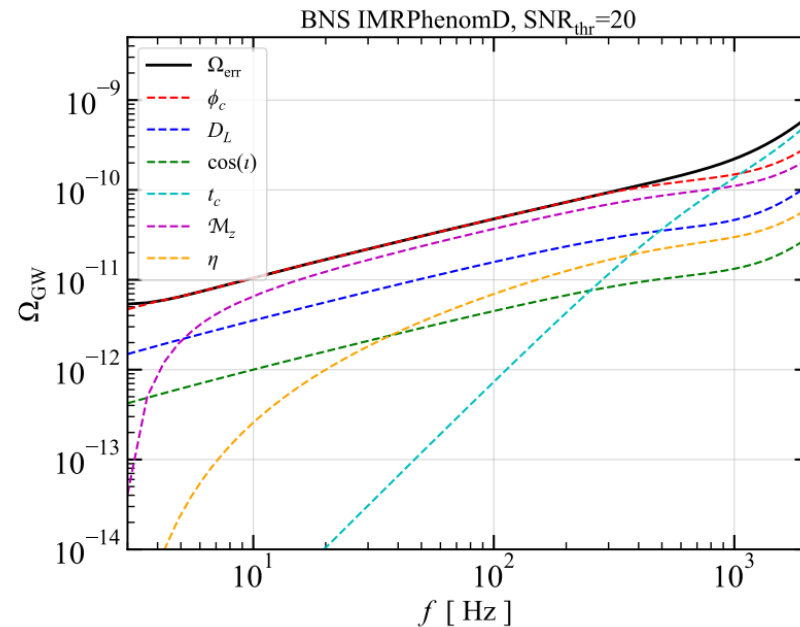
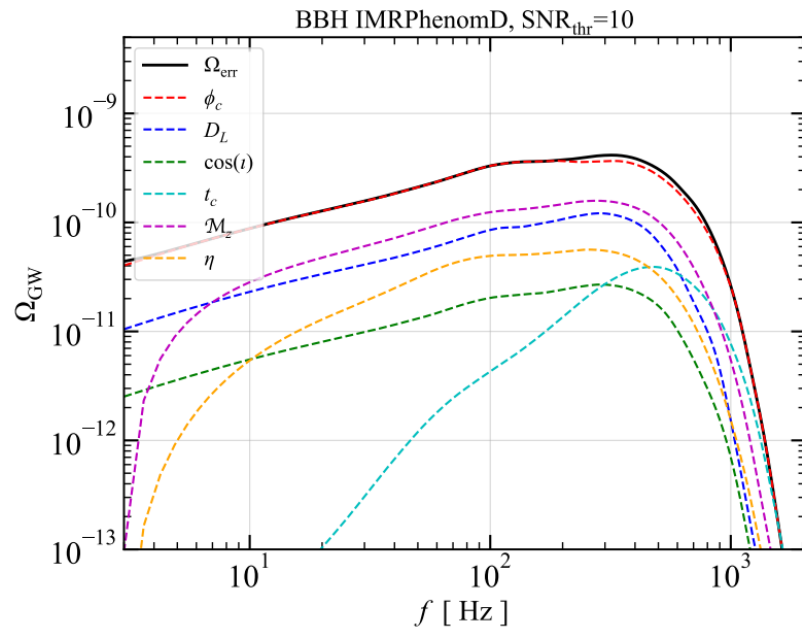
Results



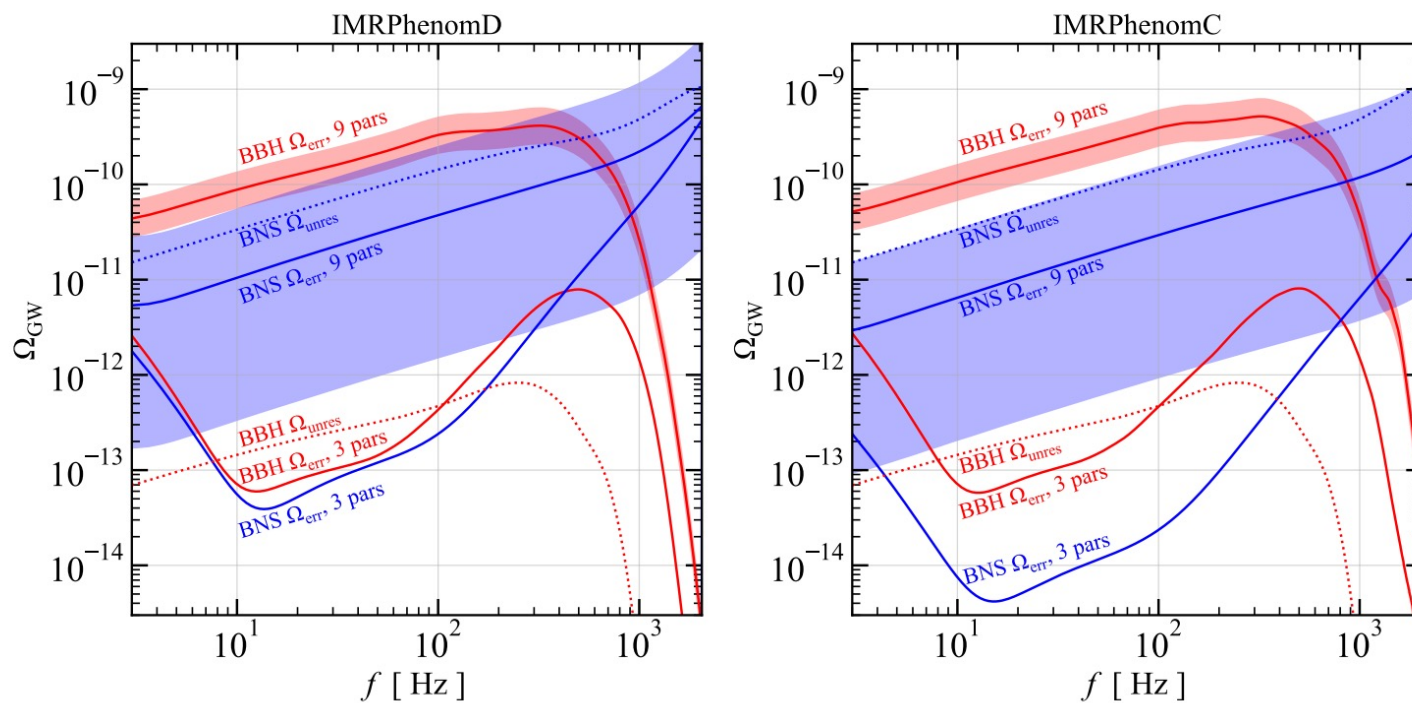
Why the large increase in foreground estimation?



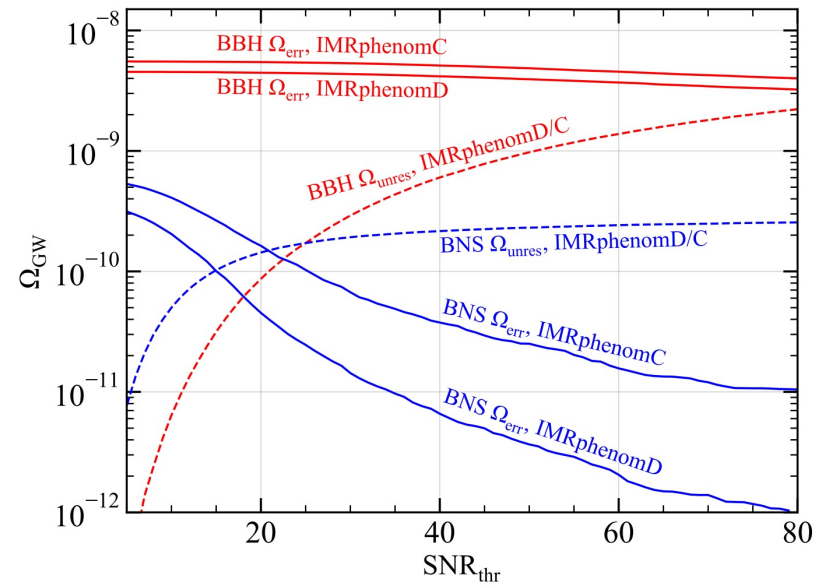
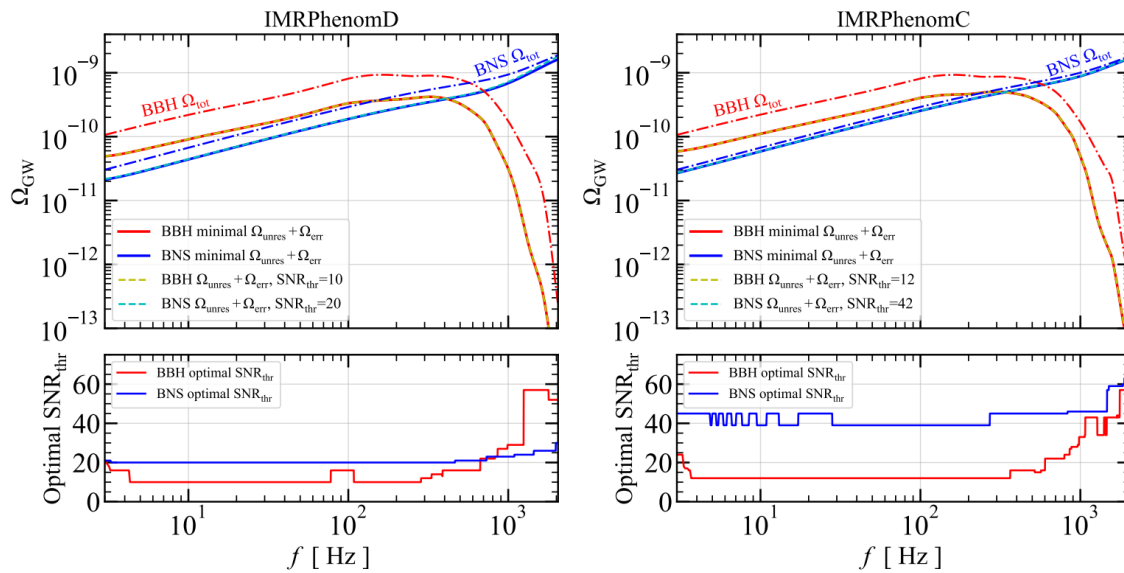
What parameters are driving this increase?



What is the split between unres and err?



Why are there waveform systematics?



Caveats and Other Subtraction Methods

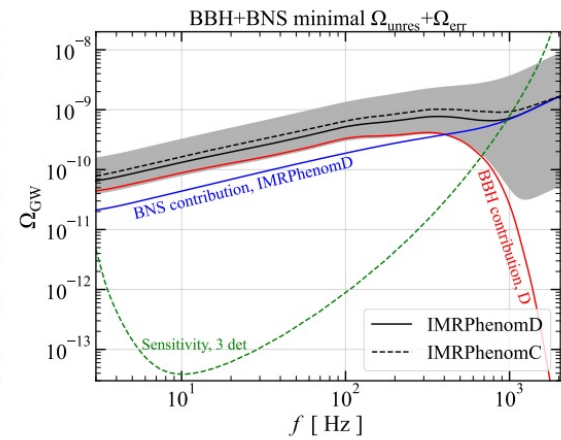
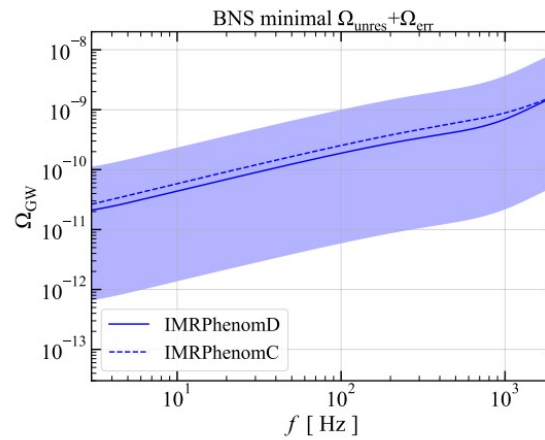
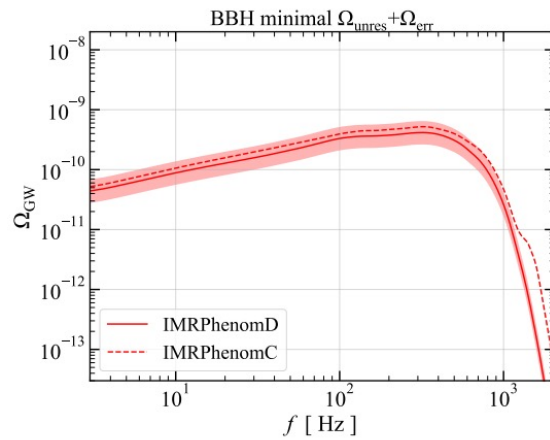
1. Multiple sources [Reali et al, 2209.13452]
2. Other Astrophysical Uncertainties (subdominant)
3. No spin (subdominant)

1. Subtraction of component tangent to signal manifold at the point of best fit [BBO: Cutler-Harms gr-qc/0511092; XG: Sharma-Harms, 2006.16116]
2. Notching in time-frequency plane [Zhong-Ormiston-Mandic, 2209.11877]
3. Simultaneous foreground and background Bayesian characterization [Smith-Thrane, 1712.00688; Biscoveanu+, 2009.04418]

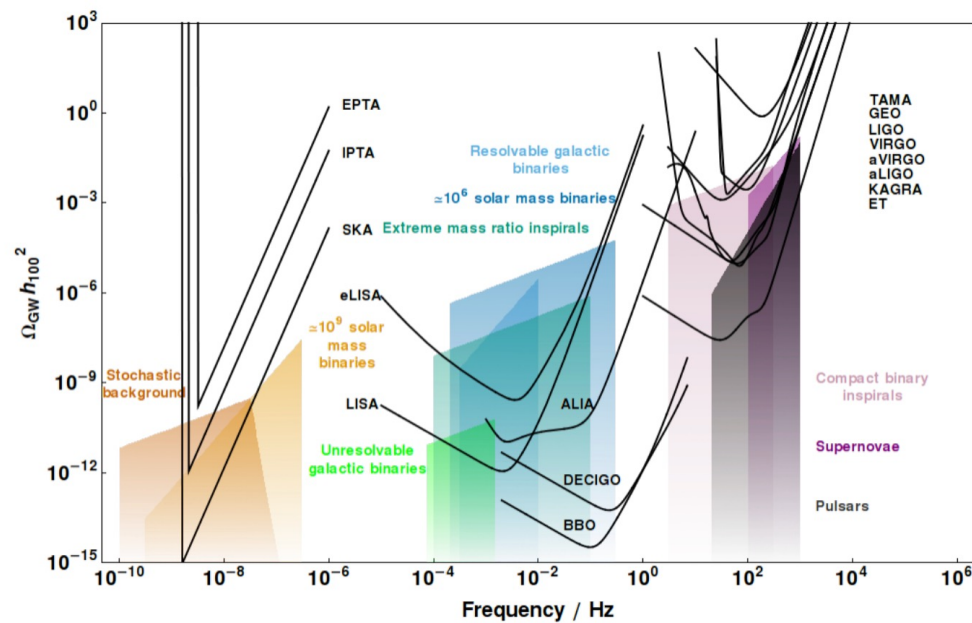
Conclusions

1. The straightforward “detect and subtract” method does not seem to work for the detection of subdominant stochastic GWs in XG detectors as there exists a large astrophysical foreground
2. Waveform systematics will become an issue once astrophysical uncertainties are reduced
3. Alternative methods of subtraction must be explored in the context of XG detectors

Extra Slides



Extra Slides



Extra Slides

Parameter	BNS	BBH
m_1	Double Gaussian [50]	POWER+PEAK [9]
m_2	Uniform [1.14, 1.46] $M_\odot$	
χ_1, χ_2	0	
z	SFR [51] + time delay	SFR [51] + time delay + metallicity
$\cos \iota$ $\cos \delta$	Uniform in $[-1, 1]$	
α, ψ, ϕ_c	Uniform in $[0, 2\pi]$	
t_c	0	